Vaishnav Mankar

Pune, Maharashtra | +91 9146790431 | vaishnavmankar2002@gmail.com
LinkedIn | GitHub | LeetCode URL

Professional Summary

M.Sc. in Scientific Computing student with a robust foundation in machine learning and data analysis using Python, Pandas, and Scikit-learn. Actively developing expertise in Deep Learning and Generative AI, with a passion for building full-stack data applications. Eager to apply strong computational and analytical abilities to solve complex, real-world problems.

Education

- **Savitribai Phule Pune University (SPPU)** — Pune, Maharashtra
  M.Sc. in Scientific Computing — Jun 2024 - Present (Expected May 2026)
  CGPA: 6.55
- **Hislop College** — Nagpur, Maharashtra
  Bachelor of Science (B.Sc.) — Jun 2020 - May 2024
  CGPA: 6.34

Technical Skills

- **Languages:** Python (Advanced), R, Java, C++, SQL
- **Data Science & ML/DL:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Statistical Modeling, PySpark
- **Deep Learning & AI:** PyTorch, TensorFlow, Generative AI, Agentic AI, LangChain, Computer Vision (OpenCV)
- **Web & Cloud:** Flask, FastAPI, Docker, AWS (Basic)
- **Databases & Tools:** MySQL, Git, GitHub, VS Code, Jupyter Notebook, Power BI

Projects

Algonix: Machine Learning Library — Jan 2025 - May 2025

Technologies: Python, NumPy, Pandas, Matplotlib, PyTest

- Developed a comprehensive machine learning library from scratch to build, analyze, and visualize transparent algorithm implementations.
- Implemented a diverse suite of supervised and unsupervised algorithms, including Linear/Logistic Regression, K-Means Clustering, and Decision Trees.
- Engineered advanced ensemble models like Gradient Boosting and incorporated optimization techniques such as Stochastic Gradient Descent.
- Designed the library with a modular architecture and published it on the Python Package Index (PyPI), making it installable via `pip`.

Real-Time Face Detection Model — May 2025 - Jun 2025

Technologies: Python, OpenCV, TensorFlow/Keras, NumPy

- Developed a robust computer vision system for real-time face localization, utilizing a pre-trained Deep Learning model for high accuracy and efficiency.
- Integrated the model with the OpenCV library to process live video streams and display bounding boxes with minimal latency.
- Optimized the pipeline for performance, achieving reliable face detection across varying camera angles, lighting conditions, and image scales.
- Code available at: GitHub Link

Road Accident Analysis Dashboard — Oct 2024 - Nov 2024

Technologies: Power BI, DAX

- Developed a dynamic Power BI dashboard to analyze and visualize complex UK road accident data, focusing on casualties, severity, and vehicle types.
- Extracted actionable insights by comparing urban and rural accident trends, providing a tool for organizations to improve road safety strategies.

Certifications

- **Data Analytics and Visualization Job Simulation** | Accenture
- **Data Science Methodology** | Cognitive Class